

## ELY, NV, USA

WMO: 724860

Lat: 39.295N

Lon: 114.847W

Elev: 1909

StdP: 80.40

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23154

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -19.9      | -17.0 | -23.3                         | 0.6 | -14.1 | -20.8 | 0.7 | -11.6 | 12.6                  | 2.4  | 11.3 | 1.3  | 4.8                   | 170  | 0.679 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.7                   | 33.1            | 13.6 | 31.9 | 13.2 | 30.7 | 12.8 | 15.7                | 26.8 | 15.0 | 26.5 | 14.3 | 26.0 | 5.2                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 12.8                            | 11.7 | 17.1 | 11.4 | 10.6 | 17.4 | 9.9 | 9.6 | 17.5 | 50.5          | 26.3 | 48.1 | 25.9 | 45.8 | 26.0 | 20.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.6          | 10.0 | 8.6 | DB                         | -25.2 | 35.5               | 3.1 | 1.1                                                | -27.4 | 36.3       | -29.2 | 37.0       | -31.0 | 37.6       | -33.3 | 38.4 | (4) |
| (5)               |      |     | WB                         | -25.4 | 17.3               | 3.1 | 1.0                                                | -27.6 | 18.0       | -29.4 | 18.6       | -31.2 | 19.1       | -33.4 | 19.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.6         | -3.1  | -1.5  | 2.9   | 5.8   | 10.3  | 16.1  | 20.5  | 19.4  | 14.4  | 7.6   | 1.7   | -3.1  | (6)  |  |
|                                                                        | DBStd       | 9.27        | 5.36  | 5.11  | 4.48  | 4.18  | 4.17  | 3.89  | 2.22  | 2.31  | 3.80  | 4.32  | 5.17  | 5.35  | (7)  |  |
|                                                                        | HDD10.0     | 1901        | 406   | 323   | 221   | 134   | 48    | 6     | 0     | 0     | 10    | 96    | 251   | 407   | (8)  |  |
|                                                                        | HDD18.3     | 4046        | 664   | 556   | 477   | 376   | 249   | 83    | 7     | 14    | 123   | 332   | 500   | 666   | (9)  |  |
|                                                                        | CDD10.0     | 1042        | 0     | 0     | 2     | 9     | 58    | 189   | 326   | 293   | 142   | 22    | 1     | 0     | (10) |  |
|                                                                        | CDD18.3     | 145         | 0     | 0     | 0     | 0     | 1     | 16    | 74    | 49    | 5     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 4557        | 0     | 0     | 0     | 6     | 121   | 874   | 1774  | 1366  | 393   | 22    | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 1813        | 0     | 0     | 0     | 1     | 23    | 322   | 831   | 554   | 81    | 1     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.2         | 4.0   | 4.1   | 4.3   | 4.4   | 4.2   | 4.4   | 4.1   | 4.2   | 4.2   | 4.1   | 4.0   | 4.0   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 248         | 18    | 19    | 23    | 25    | 28    | 19    | 17    | 21    | 21    | 21    | 17    | 18    | (15) |  |
|                                                                        | PrecMax     | 394         | 54    | 69    | 55    | 87    | 83    | 90    | 58    | 73    | 127   | 59    | 46    | 85    | (16) |  |
|                                                                        | PrecMin     | 107         | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 66          | 13    | 14    | 14    | 17    | 22    | 21    | 15    | 18    | 23    | 14    | 12    | 15    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.3  | 15.5  | 20.5  | 24.0  | 29.4  | 33.6  | 35.4  | 33.5  | 30.7  | 25.9  | 19.8  | 13.0  | (19) |  |
|                                                                        |             | MCWB        | 4.2   | 5.4   | 7.0   | 9.1   | 11.9  | 13.8  | 14.7  | 13.7  | 12.2  | 10.3  | 7.1   | 3.9   | (20) |  |
|                                                                        | 2%          | DB          | 10.3  | 12.1  | 17.5  | 21.0  | 26.5  | 31.7  | 33.8  | 32.3  | 28.8  | 23.3  | 17.3  | 10.3  | (21) |  |
|                                                                        |             | MCWB        | 3.1   | 4.0   | 5.8   | 7.6   | 10.4  | 12.5  | 14.0  | 13.4  | 11.6  | 9.0   | 6.2   | 3.2   | (22) |  |
|                                                                        | 5%          | DB          | 7.8   | 9.4   | 15.2  | 18.9  | 24.2  | 30.2  | 32.5  | 31.2  | 27.4  | 21.2  | 14.5  | 7.9   | (23) |  |
|                                                                        |             | MCWB        | 2.2   | 2.9   | 5.1   | 6.7   | 9.5   | 11.9  | 13.5  | 13.1  | 11.2  | 8.1   | 4.9   | 2.1   | (24) |  |
|                                                                        | 10%         | DB          | 5.2   | 6.7   | 12.8  | 16.4  | 22.0  | 28.6  | 31.3  | 30.0  | 25.9  | 18.9  | 11.4  | 5.5   | (25) |  |
|                                                                        |             | MCWB        | 1.0   | 1.7   | 4.2   | 5.8   | 8.7   | 11.4  | 13.1  | 12.6  | 10.8  | 7.1   | 3.6   | 0.9   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 5.6   | 5.9   | 7.7   | 9.6   | 12.9  | 14.8  | 16.7  | 17.2  | 15.2  | 11.3  | 8.2   | 5.7   | (27) |  |
|                                                                        |             | MCDB        | 10.6  | 12.7  | 18.3  | 22.5  | 26.3  | 30.2  | 27.7  | 27.2  | 24.0  | 21.7  | 16.7  | 10.8  | (28) |  |
|                                                                        | 2%          | WB          | 3.9   | 4.7   | 6.5   | 8.3   | 11.3  | 13.7  | 15.8  | 15.6  | 13.9  | 10.0  | 6.7   | 4.0   | (29) |  |
|                                                                        |             | MCDB        | 8.5   | 11.0  | 16.3  | 19.3  | 23.8  | 28.5  | 27.3  | 26.1  | 23.2  | 20.6  | 15.9  | 8.5   | (30) |  |
|                                                                        | 5%          | WB          | 2.6   | 3.3   | 5.5   | 7.2   | 10.2  | 12.8  | 15.3  | 14.8  | 12.8  | 8.9   | 5.3   | 2.6   | (31) |  |
|                                                                        |             | MCDB        | 6.9   | 8.5   | 14.3  | 17.1  | 22.1  | 27.3  | 26.9  | 25.9  | 22.8  | 18.7  | 13.5  | 7.1   | (32) |  |
|                                                                        | 10%         | WB          | 1.3   | 2.0   | 4.4   | 6.2   | 9.3   | 11.9  | 14.6  | 14.1  | 11.7  | 7.8   | 4.0   | 1.2   | (33) |  |
|                                                                        |             | MCDB        | 4.9   | 6.0   | 12.1  | 15.3  | 20.4  | 26.3  | 26.5  | 25.3  | 22.2  | 17.3  | 10.7  | 5.2   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 14.3  | 13.7  | 15.1  | 15.7  | 17.8  | 20.5  | 20.7  | 20.3  | 19.9  | 18.4  | 16.2  | 14.4  | (35) |  |
|                                                                        |             | MCDBR       | 16.9  | 16.5  | 19.2  | 21.0  | 22.3  | 23.7  | 23.2  | 22.5  | 22.3  | 22.3  | 20.6  | 16.6  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 10.7  | 9.6   | 9.8   | 10.2  | 10.1  | 10.4  | 9.5   | 9.5   | 10.0  | 11.3  | 11.4  | 10.6  | (37) |  |
|                                                                        |             | MCWBR       | 14.8  | 14.4  | 17.8  | 18.7  | 20.4  | 21.6  | 18.9  | 18.2  | 17.9  | 19.4  | 18.7  | 14.8  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.234 | 0.243 | 0.252 | 0.279 | 0.294 | 0.292 | 0.317 | 0.314 | 0.276 | 0.264 | 0.239 | 0.234 | (40) |  |
|                                                                        |             | taud        | 2.610 | 2.602 | 2.619 | 2.523 | 2.520 | 2.567 | 2.520 | 2.508 | 2.617 | 2.622 | 2.657 | 2.616 | (41) |  |
|                                                                        | Ebn at Noon |             | 975   | 1006  | 1021  | 1000  | 983   | 981   | 953   | 951   | 979   | 964   | 963   | 951   | (42) |  |
|                                                                        |             | Edh at Noon | 67    | 78    | 86    | 101   | 104   | 99    | 103   | 102   | 85    | 76    | 64    | 62    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2.48  | 3.35  | 4.73  | 5.90  | 6.75  | 7.92  | 7.10  | 6.60  | 5.65  | 4.08  | 2.82  | 2.10  | (44) |  |
|                                                                        | RadStd      |             | 0.23  | 0.35  | 0.28  | 0.40  | 0.68  | 0.61  | 0.46  | 0.47  | 0.33  | 0.30  | 0.15  | 0.22  | (45) |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) |  | N/A   | N/A     | N/A    | +0.62   | N/A   | N/A   | N/A         | N/A     | +126    | +64     |

Nomenclature: See separate page